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Reflective Report

Working through the process of training, converting, and deploying an MNIST model to a simulated edge environment gave me a better understanding of both the workflow and the practical applications behind real-world AI deployments on the edge. Even though the project was straightforward, the challenges I encountered shaped what I learned.

One of the main challenges I faced involved debugging small coding issues. Typos, syntax errors, and misplaced characters caused the code to fail at various stages, especially when loading or converting the model. These errors forced me to slow down, and read the traceback carefully, and verify each line of code. While they were frustrating in the moment, they helped me reinforce the importance of precision and attention to detail in machine learning workflows. Another challenge came from the model saving format. I initially attempted to save the trained model as a .h5 file, only to realize this format is now outdated for certain TensorFlow ops. Switching to. keras format resolved the warning message and taught me to stay aware of evolving libraries and standards.

Through this process I learned how TensorFlow Lite fits in with the broader system of AI development. Converting a full TensorFlow model into a lightweight TFLite version showed how much optimization is required for models to run efficiently on edge devices. I gained a deeper understanding of how TFLite performs on edge devices, reducing computational load and memory usage while still maintaining strong performance.

Overall, this project showed me how the technical steps of model conversion and inference connect directly to real-world constraints. TFLite enables AI to run on phone, IoT devices, embedded systems, smart cars, and other hardware where full TensorFlow models would be too heavy. Seeing the MNIST model run through the TFLite Interpreter made the concept understandable. The same model that runs on a powerful machine can be optimized to operate in environments with limited resources. This experience grew my understanding of edge AI and highlighted the important of designing models with deployment in mind from the beginning.